

LEDGER · 2026-08-28 · ENVIRONMENTAL ENGINEERING / NANOTECHNOLOGY

Biochar-Supported Sulfidated Nanoscale Zero-Valent Iron (S-nZVI@BC) for Rapid TCE Groundwater Remediation

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Pump-and-Treat Systems (with Activated Carbon)
Measured advantage	>10x acceleration in TCE chemical degradation rate constant (Kobs) in lab/column studies compared to multi-decade pump-and-treat site operation timelines
Time to first revenue	24 months
Regulatory risk	Moderate — qualification costs reported (\$1,500,000 estimated)

Why it failed

- Voided by: Mismatched comparison

Assessment

The venture fails like-for-like form factor by comparing lab chemical degradation kinetics (Kobs) of an in-situ material composite directly against the multi-decade operating

timeline of an ex-situ mechanical pumping plant. In addition, site-specific hydrogeologic permitting and field injection validation delay first paid delivery well beyond 18 months.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Duplicate references inflating the evidence base
- no_production_runsheet

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-28 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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Method

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